M4010104 / M4020104: PROGRAMMING IN PYTHON · DIGITAL UNIVERSITY KERALA · CAPSTONE PROJECT SPECIFICATION

Project 12: CPCB Multi-Pollutant Air Quality Index (AQI) Imputation & Seasonal Forecaster

Effort Budget: 10–12 Hours | Core Stack: Pure Python, NumPy, Pandas, Matplotlib, Seaborn, Plotly | Max Marks: 25 | Instructor: Dr. Sunil TT

1. Executive Overview & Domain Context

The Central Pollution Control Board (CPCB) operates continuous ambient air monitoring stations across India, tracking key criteria pollutants: PM2.5, PM10, NO2, NH3, SO2, CO, and Ozone (O3). Rapid urbanization and meteorological inversions cause severe seasonal smog episodes in winter, contrasted with dramatic monsoon washout cleaning.

Your team will build an industrial-strength environmental analytics package in Python that parses multi-city CPCB daily logs (including Kerala stations in Kochi and Thiruvananthapuram), handles intermittent sensor dropouts through seasonal median imputation, models pollutant-to-AQI regression equations in NumPy, and delivers interactive visualizers answering the 12 core questions below.

Primary Public Dataset & Telemetry/Data Schema

The project utilizes the verified public Air Quality Data in India (CPCB Stations) on Kaggle (**`https://www.kaggle.com/datasets/rohanrao/air-quality-data-in-india`**). The dataset structure captures:

- **Raw Feature Attributes:** **`City`**, **`Date`**, **`PM2.5`**, **`PM10`**, **`NO2`**, **`NH3`**, **`SO2`**, **`CO`**, **`O3`**, **`AQI`**, **`AQI_Bucket`**.
- **Engineered / Target Variables:** **`PM_Ratio`** (PM2.5 / PM10), **`Monsoon_Washout_Pct`**, **`Clean_AQI`** (imputed score), **`Pollution_Severity_Tier`**.

2. Complete Project Directory & File Architecture

Each team must organize their repository into a standardized, production-grade layout. All application source code must reside inside a dedicated package (**`src/`**), with separate directories for raw/curated data, interactive development notebooks, visual figures, and formal documentation:

```
cpcb-air-quality-forecaster/
├── README.md                          # Repository documentation & GitHub Pages source
├── requirements.txt                   # Dependency specification (numpy, pandas, matplotlib, seaborn, p
├── run_pipeline.py                    # Standalone production CLI application entry point
├── data/
│   ├── raw/
│   │   └── cpcb_city_day_raw.csv            # Raw uncleaned input dataset (including simulated/injected
│   └── processed/
│       ├── clean_cpcb_city_day.csv          # Curated, validated clean CSV dataset
│       └── cpcb_ingestion_audit.log           # Timestamped audit report detailing filtered/rejected re
├── src/
│   ├── __init__.py                    # Python package initialization
│   ├── ingestion.py                   # Defensive line-by-line file reader, range auditor & CSV exporte
│   ├── analyzer.py                    # CityAirQualityAuditor OOP class, NumPy vectorization & Pandas a
│   └── visualizer.py                  # Matplotlib envelope plots, Seaborn heatmaps & Plotly HTML expor
├── notebooks/
│   └── development_cycle.ipynb        # Development cycle notebook: EDA, trials, tests & unit assertion
└── reports/
    ├── figures/
    │   ├── primary_envelope.png       # Matplotlib confidence envelope & regression overlay
    │   ├── correlation_heatmap.png    # Seaborn annotated feature correlation heatmap
    │   └── distribution_regimes.png   # Seaborn distribution across categorical/domain tiers
```

```
├── interactive_dashboard.html     # Standalone interactive Plotly HTML dashboard with controls
└── Project_Report.pdf             # 3-4 page final technical report
```

3. Dual Development Deliverables: Notebook vs. Standalone Program

To mirror industry software engineering standards, each student group must deliver two distinct, complementary implementations of their solution:

Deliverable A: The Development Cycle Notebook (`notebooks/development_cycle.ipynb`)

The development notebook chronicles your team's engineering journey from raw data exploration to validated model prototypes. It is not merely a final script pasted into cells, but an authentic record of the development cycle:

- **Phase 1: Exploratory Data Ingestion & Diagnostics:** Loading raw data, inspecting missing values, testing boundary rules, and logging rejected records.
- **Phase 2: Mathematical Formulation & Feature Verification:** Developing and validating NumPy array calculations against target benchmarks using `np.allclose()`.
- **Phase 3: OOP Class Architecture & Unit Tests:** Instantiating the `CityAirQualityAuditor` class, executing automated unit assertions (e.g., verifying that passing invalid parameters raises an explicit `ValueError`), and testing edge cases.
- **Phase 4: Statistical Modeling & Regimen Exploration:** Implementing ordinary least-squares regression, rolling-window calculations, and multi-metric Pandas grouping.
- **Phase 5: Visual Prototype Development:** Iteratively designing and refining Matplotlib and Seaborn figures, followed by configuring the interactive Plotly HTML dashboard.

Deliverable B: The Final Standalone Python Program (`run_pipeline.py` & `src/`)

While the notebook documents the exploratory development cycle, your repository must also provide a production-ready, fully executable Python application. The program must run end-to-end directly from the terminal without requiring Jupyter:

- **Command-Line Interface (CLI):** Executed via `python run_pipeline.py --input data/raw/cpcb_city_day_raw.csv --output-dir data/processed --report-dir reports`.
- **Modular Package Import:** The driver script imports cleanly separated domain logic from `src.ingestion`, `src.analyzer`, and `src.visualizer`.
- **Terminal Feedback & Exit Codes:** The program must print structured progress indicators, log clean/faulty record counts to the console, and return an exit code of `0` upon successful pipeline completion.
- **Automated Artifact Generation:** In a single execution pass, the script reads raw inputs, writes `data/processed/clean_cpcb_city_day.csv`, saves `data/processed/cpcb_ingestion_audit.log`, exports static figures to `reports/figures/`, and generates the standalone `reports/interactive_dashboard.html`.

4. Technical Investigation Requirements & Research Questions (Q1–Q12)

Your team's implementation across the development notebook and standalone program must rigorously answer the following 12 course-aligned engineering questions:

Course Topic & Script	Q#	Concrete Engineering & Analytical Task	Python Libraries & Methods
File I/O & Exceptions (`src/ingestion.py`)	Q1	**City Timestamp & Negative Pollutant Audit:** Parse raw CPCB CSV rows using try-except. Validate that pollutant values are non-negative, filter invalid timestamps, and log corrupt sensor records into an audit log.	open(), try-except (ValueError, FileNotFoundError), range checks

Course Topic & Script	Q#	Concrete Engineering & Analytical Task	Python Libraries & Methods
Data Curation & Export (`src/ingestion.py`)	**Q2**	**Curated Multi-City Air Quality Export & Audit:** Write clean records to clean_cpcb_city_day.csv while logging stations with > 50% missing data into cpcb_ingestion_audit.log.	Defensive file writing, dictionary parsing, formatted log reporting
OOP Design (`src/ingestion.py`)	**Q3**	**City Air Quality Auditor Class:** Design an encapsulated class CityAirQualityAuditor with methods load_from_csv(), compute_seasonal_aqi(city='Kochi'), and summary_report(). Ensure invalid city raises ValueError.	OOP class, __init__, encapsulation, raise ValueError
NumPy Arrays (`src/analyzer.py`)	**Q4**	**Vectorized Pollutant Imputation & Verification:** Using only raw NumPy 2D arrays, implement seasonal month-by-city median imputation for PM2.5 and PM10. Verify imputed distributions against non-missing data via np.allclose().	np.nanmedian(), np.where(), np.allclose()
NumPy Masking (`src/analyzer.py`)	**Q5**	**Severe Smog vs. Monsoon Clean Days Masking:** Flag Severe Smog days (AQI > 300) and Good Air days (AQI < 50) in NumPy. Compare coastal cities (Kochi, Thiruvananthapuram) vs. inland cities (Delhi, Lucknow) via set intersection.	np.mean(), np.std(), boolean masks (&, |), Python set
NumPy Regression (`src/analyzer.py`)	**Q6**	**Multi-Pollutant to AQI Ordinary Least-Squares Fit:** Fit an ordinary least-squares plane (AQI = b0 + b1*PM2.5 + b2*PM10 + b3*NO2) via np.linalg.lstsq. Quantify which pollutant carries the highest variance contribution to overall AQI.	np.column_stack(), np.linalg.lstsq(), matrix product (@), R2
Pandas Series (`src/analyzer.py`)	**Q7**	**Missing-Value Distribution & Sensor Dropout Audit:** Quantify missing-value percentages across all 6 criteria pollutants by city in Pandas. Contrast linear interpolation versus forward-fill strategies.	.isna().sum(), .interpolate(), .groupby('City').mean()
Pandas Groupby (`src/analyzer.py`)	**Q8**	**City-by-Season Discretization & Pivot Table:** Discretize dates into 4 seasons (Monsoon, Post-Monsoon, Winter, Summer). Build a City-by-Season pivot_table(margins=True) showing mean PM2.5 and AQI levels.	pd.cut(), .groupby().agg(), .pivot_table(margins=True)
Pandas Reshaping (`src/analyzer.py`)	**Q9**	**Wide-to-Long Pollutant Ranking via Melt:** Reshape pollutant columns (PM2.5, PM10, NO2, SO2) into tidy long format via pd.melt() and extract the top 5 most polluted dates for Kochi and Delhi.	pd.melt(), .groupby(['City', 'Pollutant']), .nlargest(5)
Matplotlib (`src/visualizer.py`)	**Q10**	**Matplotlib Annual AQI Trend & Smog Threshold:** Plot multi-year daily AQI for selected cities in Matplotlib. Shade the CPCB Severe threshold zone (AQI > 300) in red, overlaying 30-day moving averages and custom legends.	plt.subplots(), ax.plot(), ax.axhspan(), custom legends
Seaborn (`src/visualizer.py`)	**Q11**	**Seaborn Multi-Pollutant Correlation Heatmap:** Render an annotated correlation heatmap between PM2.5, PM10, NO2, CO, and AQI with sns.heatmap(annot=True). Generate a side-by-side violin plot of AQI by season.	df.corr(), sns.heatmap(), sns.violinplot(), sns.despine()

Course Topic & Script	Q#	Concrete Engineering & Analytical Task	Python Libraries & Methods
Plotly HTML (`src/visualizer.py`)	**Q12**	**Interactive City Air Quality Comparator:** Build an interactive HTML dashboard (interactive_dashboard.html) displaying city AQI comparisons with pollutant selector dropdowns and time sliders.	plotly.express.line(), custom hover tooltips, write_html()

5. Collaborative Git Workflow, GitHub Pages & README.md Specification

Best Practices: Collaborative Git Workflow & GitHub Pages Deployment from Day 1

From the very first day, the student team must adopt professional version control and documentation practices. On Day 1, one designated team member initializes the central public GitHub repository, establishes a clear branch protection rule preventing direct pushes to `main`, adds all teammates as registered **Contributors** (with write access), and grants read access to the Course Instructor (`Dr. Sunil TT`) and teaching assistants. Immediately thereafter, the team must activate **GitHub Pages** (via *Settings → Pages → Build and deployment: Deploy from a branch: main /docs or root*) so that project documentation, interactive architecture diagrams, and the live Plotly dashboard (`reports/interactive_dashboard.html`) are publicly accessible and continuously deployed throughout the development lifecycle. All feature development must follow a strict branching model (e.g., `feature/ingestion`, `feature/analysis`, `feature/visualization`): individual team members must write small, atomic, well-commented commits from their own personal GitHub accounts and merge them into `main` exclusively through formal **Pull Requests (PRs)** accompanied by peer review. Dumping entire codebases in a single, monolithic end-of-semester commit is strictly prohibited and will result in zero marks for individual Git contribution; maintaining an unbroken, transparent Git commit and PR audit trail is essential for passing the code ownership defense during the live viva-voce examination.

Repository README.md Specification Template

The root `README.md` serves as the primary technical landing page and must strictly conform to the following structural format:

````
# Project 12: CPCB Multi-Pollutant Air Quality Index (AQI) Imputation & Seasonal Forecaster
[![Course](https://img.shields.io/badge/Course-Programming%20in%20Python-blue.svg)](https://duk.ac.in)
[![Python](https://img.shields.io/badge/Python-3.11%2B-brightgreen.svg)](https://python.org)
[![License](https://img.shields.io/badge/License-MIT-yellow.svg)](LICENSE)

## 1. Project Overview & Domain Context
- **Institution:** Digital University Kerala (MTech AI / MSc AI)
- **Course:** M4010104 / M4020104 — Programming in Python | **Instructor:** Dr. Sunil TT
- **Dataset:** Air Quality Data in India (CPCB Stations) on Kaggle (https://www.kaggle.com/datasets/roha

## 2. System Architecture & Pipeline Design
[Provide an ASCII or Mermaid architecture flow diagram showing raw ingestion -> validation -> analyzer -

## 3. Installation & Environment Setup
```bash
git clone https://github.com/<your-team>/cpcb-air-quality-forecaster.git
cd cpcb-air-quality-forecaster
python -m venv venv
source venv/bin/activate  # On Windows: venv\Scripts\activate
pip install -r requirements.txt
```

## 4. Execution Instructions
### A. Running the Standalone Application
```bash
python run_pipeline.py --input data/raw/cpcb_city_day_raw.csv --output-dir data/processed --report-dir r
```
### B. Running the Development Notebook
````

```bash
jupyter notebook notebooks/development_cycle.ipynb
```

5. Key Analytical & Statistical Findings (Q1–Q12 Summary)
[Provide a concise markdown summary table detailing raw vs. clean counts, regression coefficients, anoma

6. Interactive Telemetry Dashboard
Explore the live interactive Plotly visualization hosted on GitHub Pages:
🔗 [Live Project Dashboard](https://<your-team>.github.io/cpcb-air-quality-forecaster/interactive_dashbo

7. Contributor Git Commit & Pull Request Log Table
Contributor Name	GitHub Handle	Assigned Module / Tasks	Verified Commit Hash Range	Merged Pull
[Student Full Name]	`@github_handle`	Modular Architecture (`src/ingestion.py`, `src/analyzer.py`,

6. Evaluation Rubric & Mandatory Academic Integrity Governance

Projects are evaluated on a 25-mark scale based on code quality, mathematical correctness, visualization depth, git collaboration discipline, and live technical defense:

Assessment Criterion	Basis of Evaluation & Deliverable Verification	Max Marks
1. Data Curation & File I/O Pipeline (Q1–Q2)	Robust line-by-line file reading with defensive **`try-except`** handling, physical/domain bounds validation, clean CSV output, and structured **`cpcb_ingestion_audit.log`** reporting.	**5 Marks**
2. OOP, NumPy & Pandas Analytics (Q3–Q9)	Modular **`CityAirQualityAuditor`** class architecture, vectorized NumPy feature re-derivations (verified via **`np.allclose`**), regression modeling, and Pandas rolling/groupby/melt operations.	**6 Marks**
3. Multi-Library Visualizations (Q10–Q12)	Publication-quality Matplotlib confidence envelope/trend plot, Seaborn correlation heatmap and distribution violin plot, and standalone interactive Plotly HTML dashboard.	**5 Marks**
4. Software Packaging, GitHub & Documentation	Clean directory layout, comprehensive **`README.md`**, working GitHub Pages site, **`development_cycle.ipynb`** notebook, standalone **`run_pipeline.py`**, and balanced individual Git commits/PRs.	**4 Marks**
5. Live Presentation & Code Viva-Voce Defense	Technical walkthrough and individual live viva defense where each student must explain or modify lines of code in real time without external assistance.	**5 Marks**

> **STRICT ACADEMIC INTEGRITY & ZERO-TOLERANCE AI POLICY (0 MARKS PENALTY):** All submitted Python source code (**`.py`**, **`.ipynb`**) and written documentation will be screened through an automated **AI-Code & Plagiarism Detection Suite**. If AI-generated code or synthetic documentation is detected at any checkpoint or final submission, **0 Marks will be awarded immediately** for the entire project. Each student must prove authentic individual code ownership during the live viva-voce examination — *Course Instructor: Dr. Sunil TT*.

7. Milestone Schedule, Submission Checklist & Viva Defense Prompts

To ensure systematic progress across the semester, teams must adhere to three project checkpoints and prepare for on-the-spot live code modification challenges during the viva-voce examination:

Milestone Stage	Scheduled Window	Required Deliverables & Verification Milestones
Checkpoint 1 (Data Ingestion)	Week 11 Lab Session	Repository initialized with contributors and GitHub Pages active. **`src/ingestion.py`** operational with line-by-line filtering, **`clean_cpcb_city_day.csv`** exported, and **`cpcb_ingestion_audit.log`** generated.
Checkpoint 2 (Analytics & OOP)	Week 13 Lab Session	**`src/analyzer.py`** complete with **`CityAirQualityAuditor`** class, NumPy feature checks (Q4), outlier masks (Q5), regression modeling (Q6), and exploratory notebook sections.
Final Submission & Viva	Week 15 Project Review	Full package operational: standalone **`run_pipeline.py`**, polished **`development_cycle.ipynb`**, interactive **`interactive_dashboard.html`** on GitHub Pages, 3–4 page PDF report, and live code defense.

Representative Live Viva-Voce Code Modification Challenge

During the individual code viva-voce, the instructor will ask students to execute an immediate 2-minute code modification on their working repository to demonstrate genuine authorship:

- **Challenge Prompt:** *"In src/analyzer.py, compare forward-fill vs median seasonal imputation for missing PM2.5 values in Kochi stations. Re-run your module and report the difference in annual mean AQI."*